North South University

Department of Electrical & Computer Engineering

Lab Report

Experiment No: 03

Experiment Title: Combinational Logic Design

Course Code: CSE231L

Section: 10

Course Name: Digital Logic Design Lab

Lab Group #: 07

Written By: Mohammad Tanvirul Hasan Riyad

Date of Experiment: 11/10/25

Date of Submission: 18/10/25

Group Members ID:	Group Members Name:
2413692042	Tanvirul Hasan Riyad
2322723642	Md Mafiul Haque
2411600042	Md. Tayasshuk Imam
2413054042	Tawhid Hasan

Experiment Name: Combinational Logic Design

Objectives:

1. Becoming ~~famirri~~ familirize with the analysis of combinational logic network.

2. The implementation of networks using two canonical forms.

3. Devise combination circuit using universal logic.

4. Understand basic binary arithmatic circuits.

Apparatus:

- Trainer Board
- 1XIC 4073 ~~7411~~ Tripple 3-input And gates
- 2XIC 4075 ~~7432~~ Tripple 3-input OR gates
- 1XIC 7404 Hex Inverters (NOT gates)

Theory:

Minterm for each combination of the variables that produce a 1 in the function and taking the 'OR' of all those terms.

- Minterm in variables product of n literals in which each variable appears exactly one either in T or F form.

- Each minterm has value 1 for exactly one combination of values of the variables that produce eg. ABC(111) ⇒ m7.

- A function can be written as a sum of minterms, reffered to as a minterm expansion or a standard sum of products.

Maxterm for each combination of variables that produces a 0 in the function and then taking the AND of all these terms
maxterm of n variables = sum of n literals in which each variable appears exactly one in T or F form.

- Each maxterm has a value of 0 for exactly one combination of values of variables.
(eg. $A+B+\bar{C}(001) \Rightarrow M_1$ (value is 0))

• Function can be written as a product of maxterms, which is reffered as a maxterm expansion or POSOP (Product of Sum).

Canonical forms:

The technique that is used to represent the mathematical entities or matrix in it's standard form is formed as canonical form. The term canonicalization is also known as standardization or normalization with respect to the equivalence relation.

Experimental procedure:

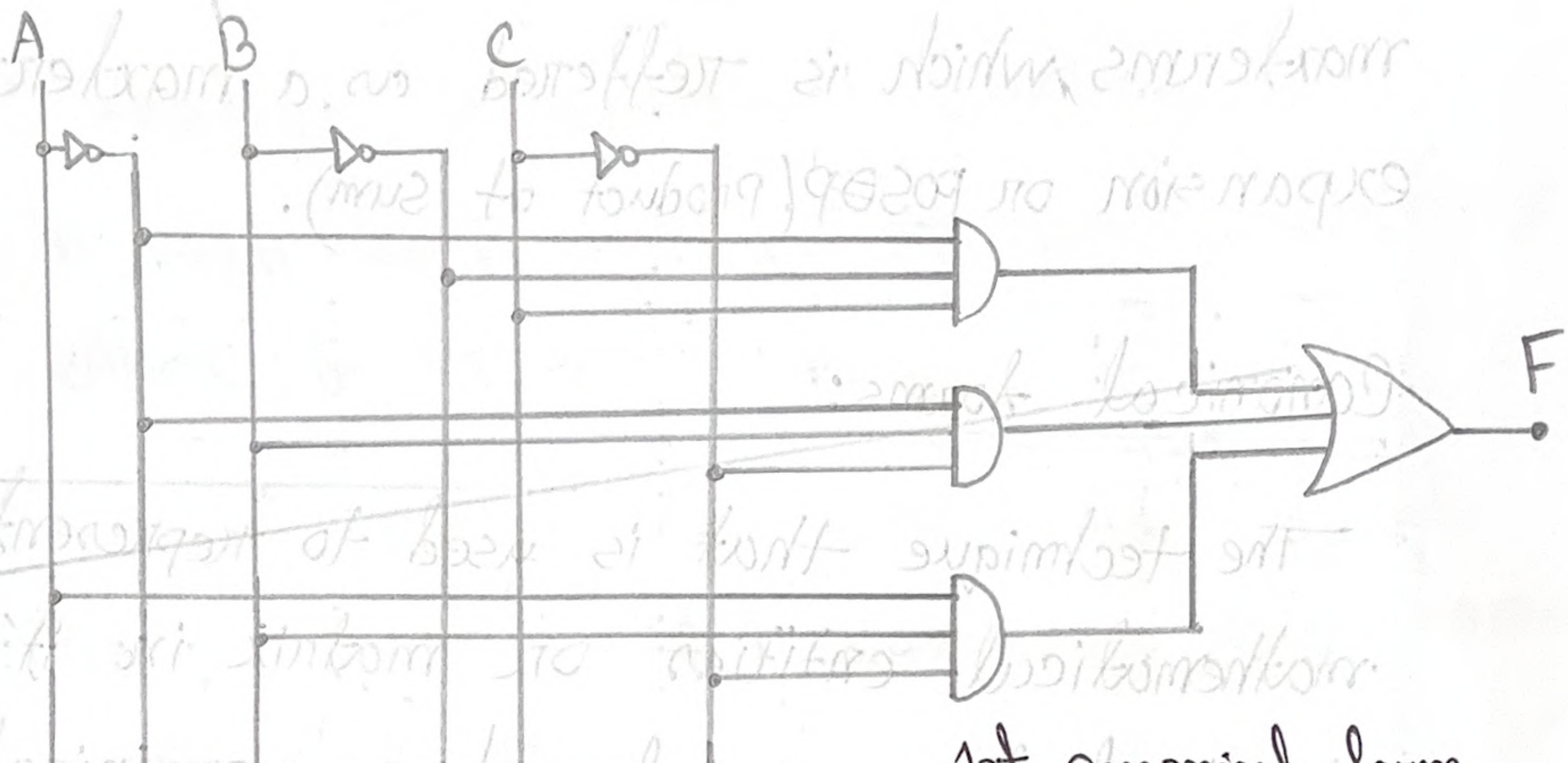
All the minterms of three inputs ABC:

$\bar{A}\bar{B}\bar{C}(m_0)$, $\bar{A}\bar{B}C(m_1)$, $\bar{A}B\bar{C}(m_2)$, $\bar{A}BC(m_3)$, $A\bar{B}\bar{C}(m_4)$, $A\bar{B}C(m_5)$, $AB\bar{C}(m_6)$, $ABC(m_7)$

All the maxterms for three inputs ABC:

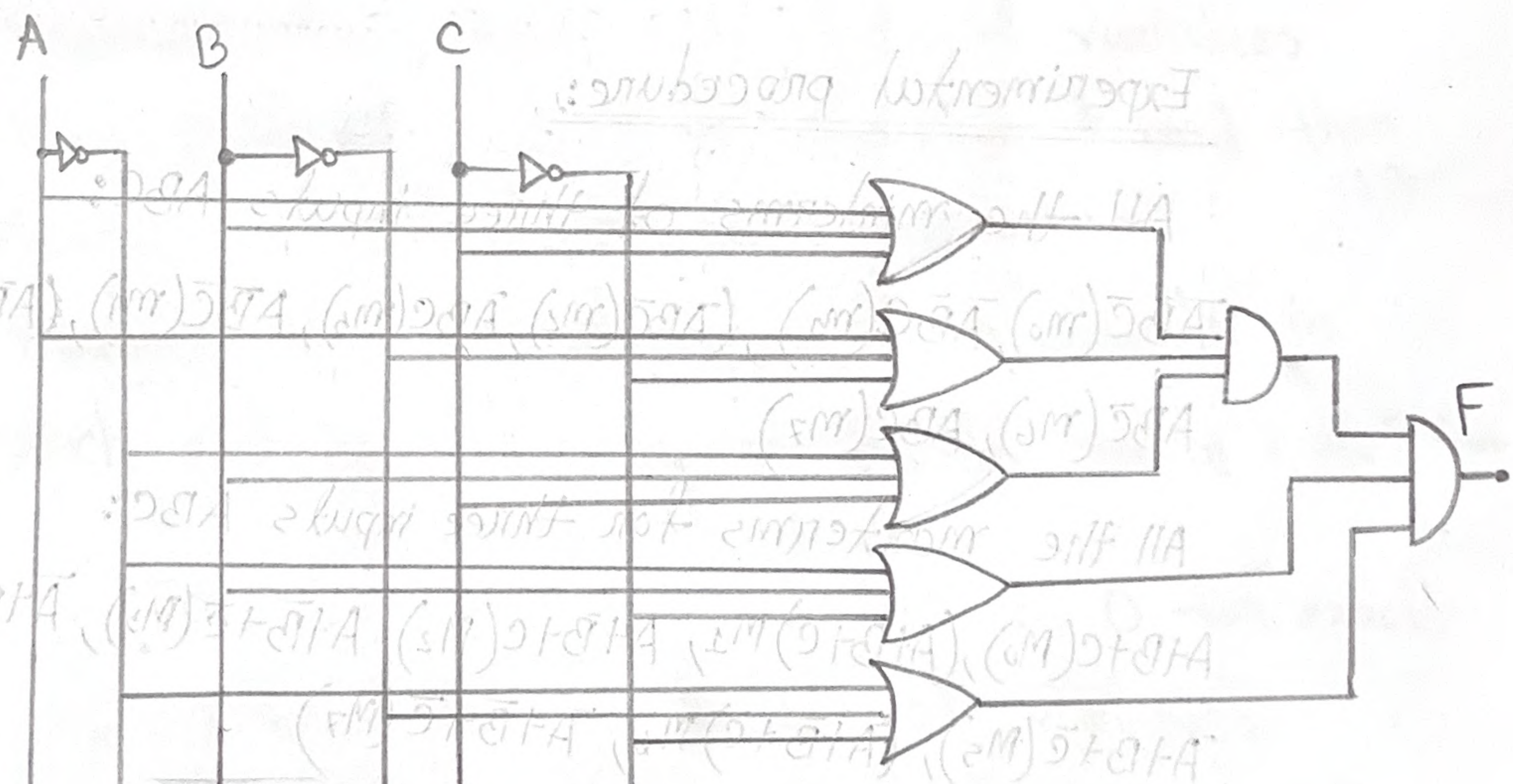
$A+B+C(M_0)$, $(A+B+\bar{C})M_1$, $A+\bar{B}+C(M_2)$, $A+\bar{B}+\bar{C}(M_3)$, $\bar{A}+B+C(M_4)$, $\bar{A}+B+\bar{C}(M_5)$, $(\bar{A}+\bar{B}+C)M_6$, $\bar{A}+\bar{B}+\bar{C}(M_7)$

Circuit Diagram:



1st canonical form

$F = \bar{A}BC + \bar{A}B\bar{C} + AB\bar{C}$



2nd canonical form

$F = (A+B+C)\cdot(A+\bar{B}+\bar{C})\cdot(\bar{A}+B+C)\cdot(A+B+\bar{C})\cdot(\bar{A}+\bar{B}+\bar{C})$

Experimental Data Table:

Input Reference	ABC	F	Minterm	Maxterm
0	000	0	$\bar{A}\bar{B}\bar{C}$ (m_0)	$A+B+C$ (M_0)
1	001	1	$\bar{A}\bar{B}C$ (m_1)	$A+B+\bar{C}$ (M_1)
2	010	1	$\bar{A}B\bar{C}$ (m_2)	$A+\bar{B}+C$ (M_2)
3	011	0	$\bar{A}BC$ (m_3)	$A+\bar{B}+\bar{C}$ (M_3)
4	100	0	$A\bar{B}\bar{C}$ (m_4)	$\bar{A}+B+C$ (M_4)
5	101	0	$A\bar{B}C$ (m_5)	$\bar{A}+B+\bar{C}$ (M_5)
6	110	1	$AB\bar{C}$ (m_6)	$\bar{A}+\bar{B}+C$ (M_6)
7	111	0	ABC (m_7)	$\bar{A}+\bar{B}+\bar{C}$ (M_7)

~~Table~~ Truth table to a combinational circuit

	Shorthand Notation	Function
1st Canonical form	$F=\Sigma(1,2,6)$	$F=\overline{A}\,\overline{B}C+\overline{A}B\overline{C}+AB\overline{C}$
2nd canonical form	$F=\pi(0,3,4,5,7)$	$F=(A+B+C)\cdot(A+\overline{B}+\overline{C})\cdot(\overline{A}+B+C)\cdot(\overline{A}+B+\overline{C})\cdot(\overline{A}+\overline{B}+\overline{C})$

1st and 2nd canonical forms of the combinational circuit

Questions and Answers:

1. A minterm is a product (AND) term that includes every variable of a function, either in its normal or complemented form. The sum-of-products canonical form is a Boolean expression written as a sum (OR) of these minterms.

The expression $F = AB' + ABC'$ is not in the first canonical form. For a function with three variables (A, B, C), every term must contain all three. The term AB' is missing the variable C.

2. No, a function in canonical form is not always its minimal form, but it can be simplified.

Example: The canonial form $F = x'y + xy$ can be simplified to $F = y$; which is its minimal form.

03.

For 4 variables (A, B, C, D), there are $2^4 = 16$ minterms and 16 maxterms.

Given:

$$Z = (A+B'+C+D') \cdot (A'+B'+C'+D) \cdot (A'+B+C+D) \cdot$$
$$(A'+B'+C'+D') \cdot (A+B+C+D') \cdot (A+B'+C'+D)$$

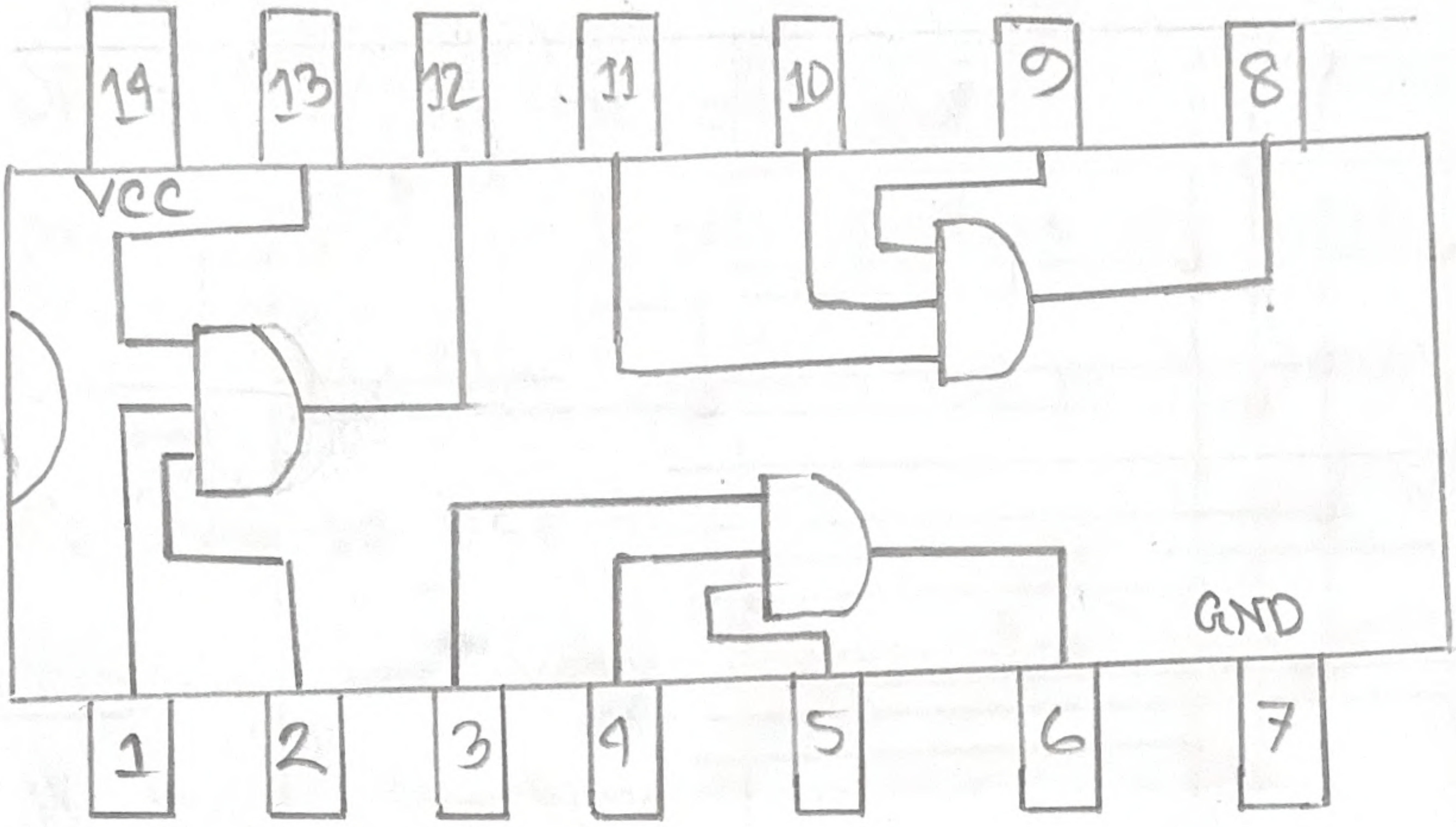
The given expression corresponds to the maxterm indices:

$$Z = \pi(1,5,6,8,13,14,15)$$

The final sum-of-minterms expression is:

$$Z = \Sigma(0,2,3,4,7,9,10,11,12)$$

04.

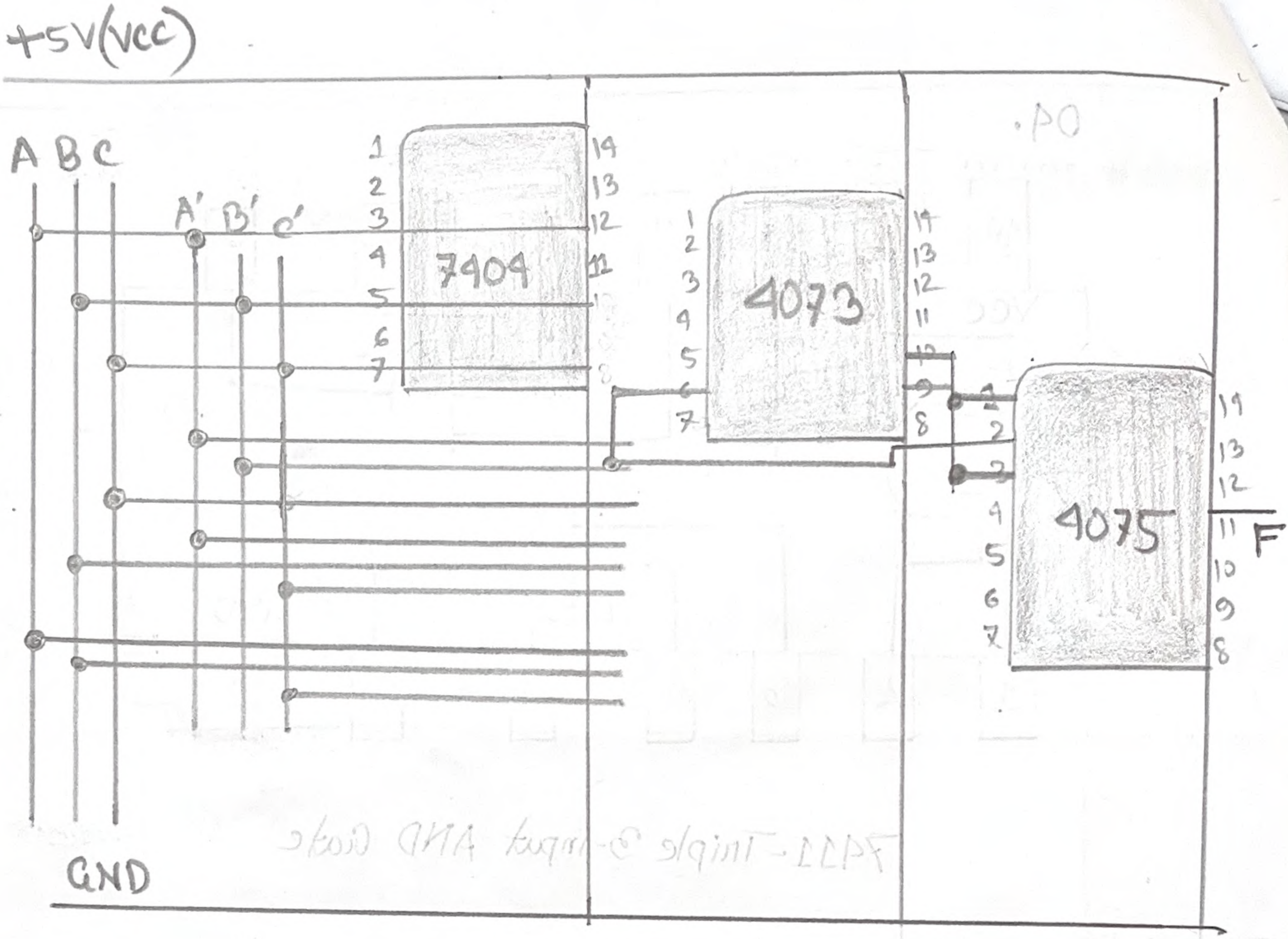


7411-Triple 3-input AND Gate

05. The function is,

$F=\Sigma(1,2,6)$ or $F= A'B'C + A'BC' + ABC'$

P.T.O.



Discussion:

In this experiment, we learned about the two canonical forms and how they works. We also learned how two different circuit can give same result if arranged properly. For this experiment, wed had 1XIC Hex Inverter, 1XIC 4073

3-input AND gate and IC with three input OR gate 4075. For minterm we didn't face any problem as it only required one AND & one OR & a Hex Inverter. But in Maxterm we had to use two OR IC gate and the circuit was very complex for that reason it took more time.

While using two OR gates we used 1 OR IC for three Maxterm then connected output to AND gate input while the two other OR output from the other OR IC. We connected these inputs to second AND input and got expected output.

While, the circuit in maxterm we needed to connect a lot of gates also we needed a lot of wires. Though it was complex our experiment were working perfectly.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Hex Digit Display
LED Matrix
TTY
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Reptar Local Bus
RGB Video
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1st Canonical: A'B'C+A'BC'+ABC'


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A
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C
2nd Canonical: (A+B+C).(A+B'+C').(A'+B+C).(A'+B+C').(A'+B'+C')

E. Experimental Data

Instructor's Signature: ....

Group: 07	Date: 11/10/25	Section : 10

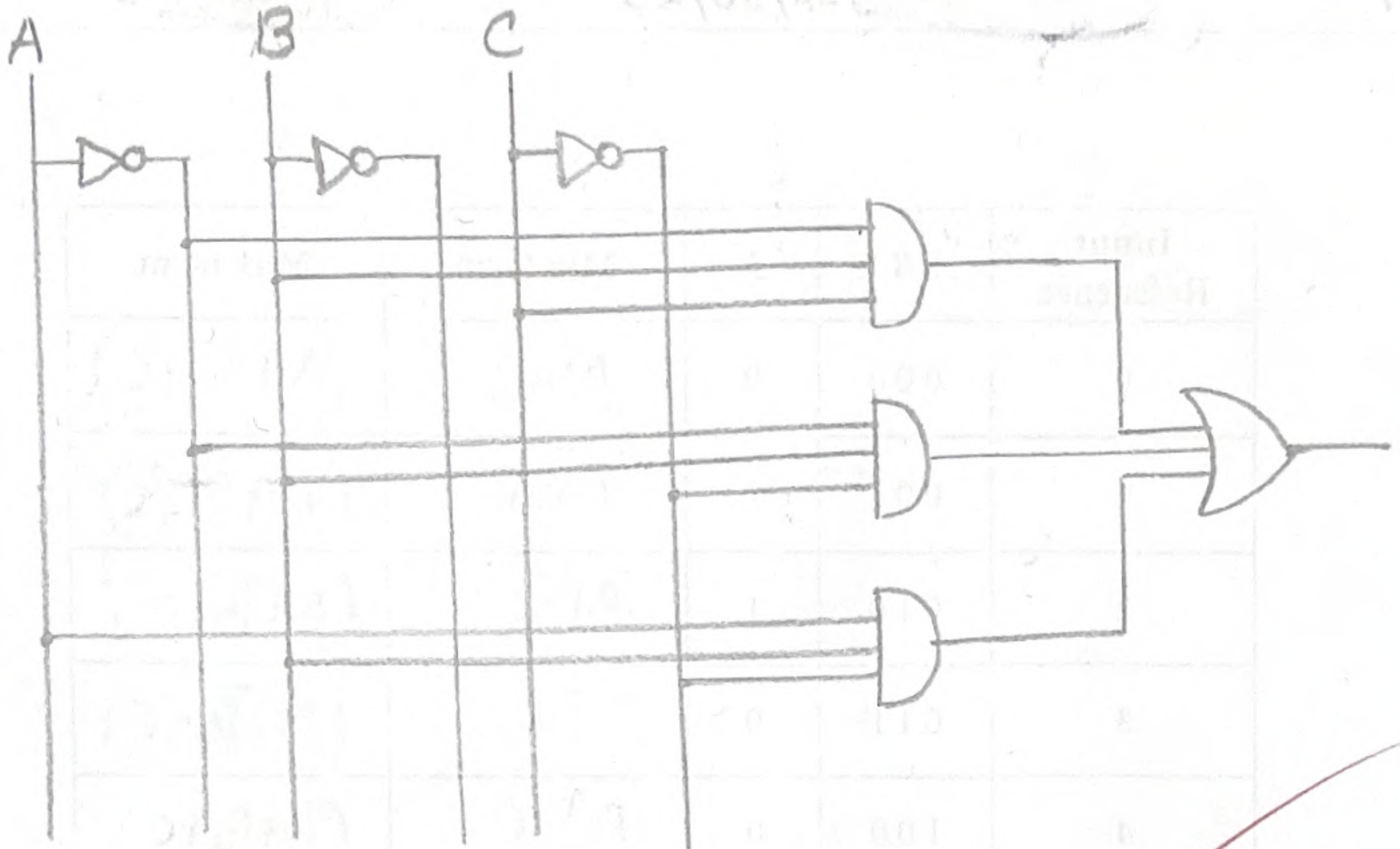
Input Reference	A B C	F	Min term	Max term
0	0 0 0	0	$\bar{A}\bar{B}\bar{C}$	$(A+B+C)$
1	0 0 1	1	$\bar{A}\bar{B}C$	$(A+B+\bar{C})$
2	0 1 0	1	$\bar{A}B\bar{C}$	$(A+\bar{B}+C)$
3	0 1 1	0	$\bar{A}BC$	$(A+\bar{B}+\bar{C})$
4	1 0 0	0	$A\bar{B}\bar{C}$	$(\bar{A}+B+C)$
5	1 0 1	0	$A\bar{B}C$	$(\bar{A}+B+\bar{C})$
6	1 1 0	1	$AB\bar{C}$	$(\bar{A}+\bar{B}+C)$
7	1 1 1	0	ABC	$(\bar{A}+\bar{B}+\bar{C})$

Table F.1 Truth table to a combinational circuit

	Shorthand Notation	Function
1st Canonical Form	$F = \Sigma(1,2,6)$	$F = \bar{A}\bar{B}C+\bar{A}B\bar{C}+AB\bar{C}$
2nd Canonical Form	$F = \Pi(0,3,4,5,7)$	$F = (A+B+C)\cdot(A+\bar{B}+\bar{C})\cdot(\bar{A}+B+C)\cdot(\bar{A}+B+\bar{C})\cdot(\bar{A}+\bar{B}+\bar{C})$

Table F.2 1st and 2nd canonical forms of the combinational circuit of Table F.1

1st Canonical Form



2nd Canonical Form

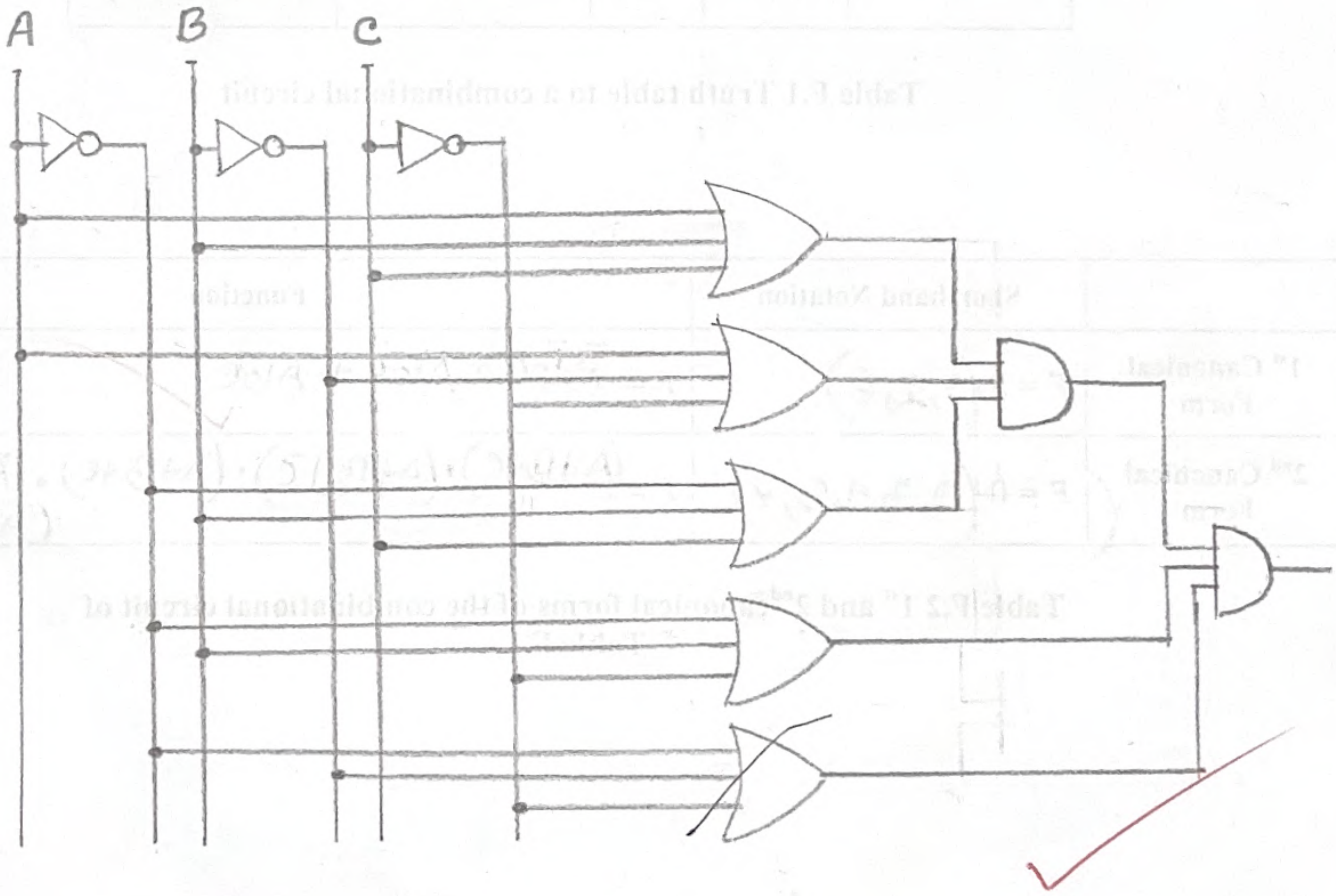


Figure F.1 1st and 2nd canonical circuit diagrams of the combinational circuit of Table F.1